CASE PRESENTATION

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PARTICULARS

- Name – Mrs. XYZ
- Age – 71 years
- Sex – Female
- Occupation – Homemaker
- Address – Arsikere, Hassan
- Socioeconomic status – Upper lower class (modified Kuppuswamy)
- Date of admission -12.04.2020
- Date of examination -12.04.2020

CHIEF COMPLAINTS

- Cough with expectoration since 2 months
- Breathlessness since 15 days
- Fever since 4 days

HISTORY OF PRESENTING ILLNESS

The patient was apparently normal 2 months ago. Then she developed cough, 5-6 episodes per day, which has gradually progressed and now it is throughout the day. It is associated with expectoration which is white in colour, mucoid in consistency, moderate amount, not blood tinged, not foul smelling, no positional variation, no diurnal variation.

She complains of breathlessness (without wheeze) since 15 days . Initially it was present while walking upstairs, it rapidly worsened and now she has breathlessness even during day to day activities.

It aggravates on lying on right side and partially relieves on lying on left side and sitting.

There is no history suggestive of orthopnea or PND.

- She developed fever since 4 days which is present throughout the day, not associated with chills or rigors. It was partially relieved on taking paracetamol.
- Loss of appetite since 2 months.
- Complains of significant weight loss (6-8kgs) in past 2 months.
- Disturbed sleep since 4 days
- She had chest pain 2 months which lasted for 5-7 days. It was left sided, catchy type , would aggravate on taking deep breath or coughing and would partially relieve on lying on left side.

- No history of any bone pain, abnormal bleeding PV , hematuria , breast lump or dysphagia.
- No history of trauma to the chest any time.

PAST HISTORY

- Has undergone hysterectomy when she was 40 years for abnormal excessive bleeding.
- She was diagnosed with kidney cancer 8 years ago when she had complaints of blood in urine and her right kidney was removed.
- She is a known case of hypertension since 8 years and is on regular medication.
- 8 months ago there was a decrease in voice for which she had consulted a doctor, CT scan was done and was told to have lesion suggestive of cancer and advised further work up . But ,as the patient was relieved of the symptom she did not go for further work up.

- She is not a known case of Diabetes , Bronchial Asthma, Epilepsy or Thyroid Disorders.
- No history of Tuberculosis in the past

DRUG HISTORY

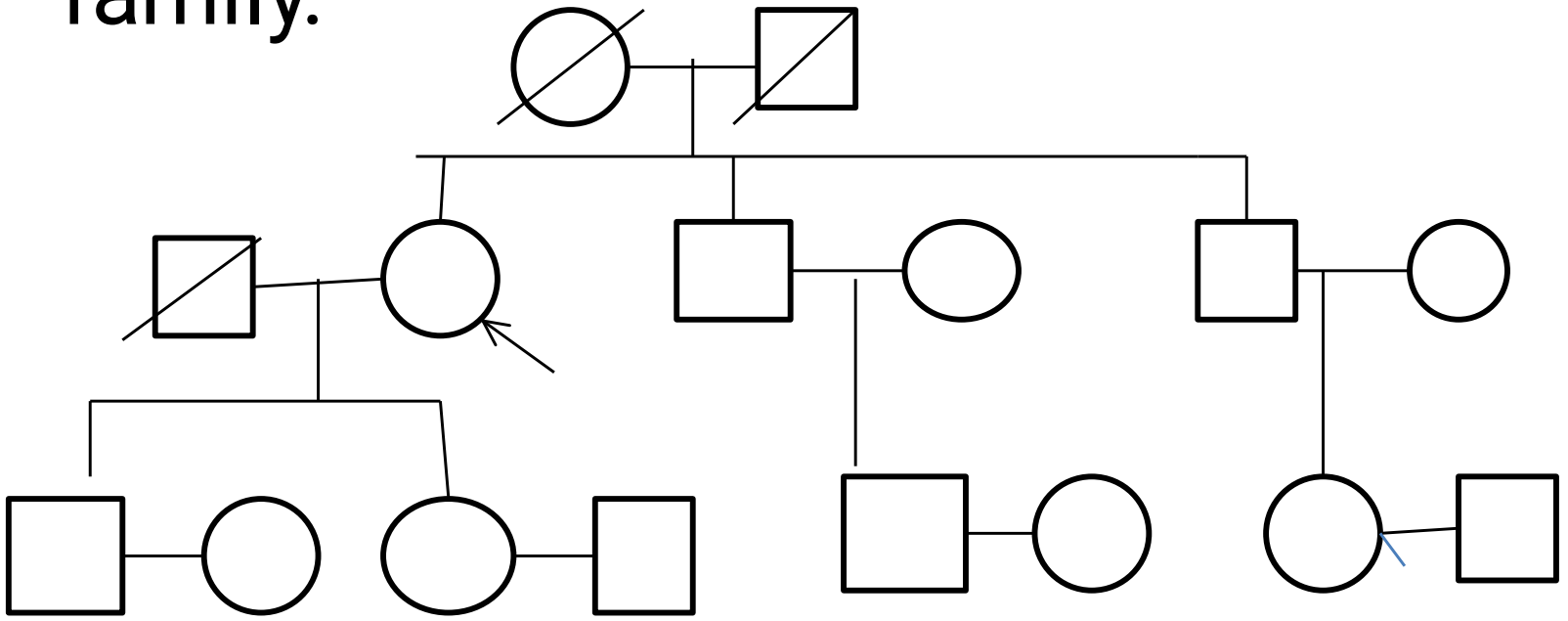
- She is not known to be allergic to any drug
- She is on regular medication for hypertension for the past 8 years (Betaloc 50mg BD)

PERSONAL HISTORY

- Diet- Mixed
- Appetite- Decreased since 2 months
- Sleep- Disturbed since 4 days
- Bowel and Bladder habits- Regular and normal
- Not a known smoker or chronic alcoholic.
- No H/O tobacco use or substance abuse.

FAMILY HISTORY

- No history of Tuberculosis, Bronchial Asthma, Malignancy in members of the family.



SUMMARY

A 71 year old lady Mrs. XYZ,

- who was operated for Renal Carcinoma 8 years ago
- hypertensive since then
- with history of pleuritic chest pain on left side 2 months ago which lasted for 5-7 days.
- presented with cough with expectoration, loss of appetite and loss of weight of 2 months duration,
- breathlessness since 15 days with rapid worsening, trepopnic in nature
- fever since 4 days

Based on the history it might be a case of left sided Pleural Effusion.

THE CAUSE MAY BE:

- Considering old age and rapid weight loss and cough it may be a case of Bronchogenic carcinoma.
- Since there is history of cough since 2 months and weight loss it may be Tuberculosis.
- Considering past history of right sided nephrectomy for renal carcinoma it may be Metastatic Carcinoma.

- ?? Synpneumonia

PHYSICAL EXAMINATION

An elderly lady, who is moderately built and nourished, conscious, co-operative and well oriented to time, place and person.

VITALS

- Pulse- 64 beats per minute, regular, good volume, normal character, all peripheral pulses are felt.
- Blood Pressure- 110/78mmHg, measured in right arm in supine position
- Respiratory rate- 28 cycles per minute , Abdominothoracic
- Acessory muscles?? / effort of breathing at rest?

- She is afebrile at the time of examination
- Weight-62kg Height-156cm BMI-25.47kg/m2
- Pallor is present
  - Koilonychia / icterus/ Petioche / prurpura/ St tenderness / generalised lymphadenopathy
- No cyanosis
- No clubbing
- Presence of subcostal scar on right side 14cm.
- Presence of transverse lower abdominal scar 10cm .
- Both the breasts appear normal
- No neck mass

EXAMINATION OF RESPIRATORY SYSTEM

UPPER RESPIRATORY TRACT

- Nasal cavity- Normal
- Oral cavity- Normal
- Pharynx appears normal

EXAMINATION OF LOWER RESPIRATORY SYSTEM

INSPECTION

- Chest is elliptical in shape and chest movements appears to be reduced on left side
- Trachea appears to be deviated to right
- Intercostal spaces on left side appear full
- Apex beat cannot be visualised
- No use of accessory muscles of respiration
- No dilated veins, scars or sinuses
- Kyphoscoliosis, drooping of shoulder not present
- No visible lymph nodes

MOVEMENTS OF CHEST WALL

AREAS OF CHEST WALL	RIGHT	LEFT
ANTERIORLY •Supraclavicular •Infraclavicular •Inframammary	Normal	Appears to be reduced
POSTERIORLY •Suprascapular •Interscapular •Infrascapular	Normal	Appears to be reduced

PALPATION

- Inspectory findings are confirmed.
- Trachea is deviated to right.
- Apex beat cannot be localized.
- No palpable lymph nodes.
- No intercostal tenderness.
- Measurements:

  AP diameter-30cm

  Transverse diameter-38cm

  Chest circumference (on expiration)-96cm

  Chest circumference (on deep inspiration)-99cm

  Chest expansion-3cm

  Right hemithorax-49cm (expansion-2.5cm)

  Left hemithorax-50cm (expansion-0.5cm)

MOVEMENTS OF CHEST WALL

AREAS OF CHEST WALL	RIGHT	LEFT
ANTERIORLY •Supraclavicular •Infraclavicular •Inframammary	Normal	Reduced
POSTERIORLY •Suprascapular •Interscapular •Infrascapular	Normal	Reduced

PERCUSSION

DIRECT

Clavicle

Right	Left
Resonant	Dull

INDIRECT

ANTERIORLY •Supraclavicular •Infraclavicular •Inframammary	•Resonant •Kronigs isthmus is resonant. •Liver dullness from 5th intercostal space in mid clavicular line	• Stony dullness • Traube's space obliterated • Kronigs isthmus is dull. • Cardiac dullness could not be made out separately
LATERALLY •Axillary •Infra-axillary	Resonant	Stony dullness
POSTERIORLY •Suprascapular •Interscapular •Infrascapular	Resonant	Stony dullness

AUSCULTATION

Areas of chest wall	Right	Left
ANTERIORLY •Supraclavicular •Infraclavicular •Mammary	Vesicular breath sounds	Absence of breath sounds
Laterally •Axillary •Infra-axillary	Vesicular breath sounds	Absence of breath sounds
Posteriorly •Suprascapular •Interscapular •Infrascapular	Vesicular breath sounds	Absence of breath sounds

- No added sounds (…)

SYSTEMIC EXAMINATION

1. Cardiovascular system
   S1 and S2 heard.
   No added sounds.
2. Abdominal examination
   soft , non tender
   Bowel sounds normally heard.
   No mass or fluid.
3. Central Nervous System
   Higher mental functions are normal
   Motor and sensory system normal.

SUMMARY

A 71 year old lady Mrs.XYZ, who was operated for renal carcinoma 8 years ago ,presented with cough with expectoration, loss of appetite and weight loss for the past 2 months, breathlessness since 15 days with rapid worsening, trepopnic in nature and fever since 4 days .

On examination

- Pallor is present,
- Trachea is deviated to right,
- Stony dullness on left side,
- Decreased movement , vocal fremitus, and vocal resonance on left side
- Absence of breath sounds on left side

PROVISONAL DIAGNOSIS

It is a case of **LEFT SIDED MASSIVE PLEURAL EFFUSION**

- **Points in favour :**

-Based on history:

1.Pleuritic chest pain

2.Trepopnea

3.Rapid worsening of breathlessness

-Based on examination

1.Trachea is deviated to the opposite side(right)

2.Intercostal space-full

3.Reduced chest movements and expansion on left side

4.Stony dullness on percussion

5. vocal resonance and vocal fremitus on left side.

- **Points against**

-Based on history

1.Cough with expectoration since 2 months

-Based on examination

1.Deviation of trachea could be made out

but mediastinal shift could not be assessed

- **Causes might be**

1.Primary Bronchogenic carcinoma

Considering age, sex, non smoking history,

rapidity of symptoms and history of abnormal CT chest 8 months ago it might

be Bronchogenic carcinoma

2.Metastatic lung disease

Considering the history of nephrectomy for Renal carcinoma and history of abnormal CT chest 8 months ago ,metastatic cause must be considered.

3. Tuberculosis.-

Secondary progressive Tuberculosis

DIFFERENTIAL DIAGNOSIS

1.FIBROTHORAX

- **Points in favour**
  1. Dull note on percussion on left side.
  2. Reduced chest movement on affected side.
  3. Diminished vesicular breath sounds with diminished vocal resonance on left side.
- **Points against**
  1. Rapidity of onset
  2. left Intercostal spaces fullness, left hemithorax more than right hemithorax.
  3. Absence of crowding of ribs with drooping of shoulder.

2.THICKENED PLEURA

- **Points in favour**
  1. Dull note on percussion
  2. Diminished vesicular breath sounds and vocal resonance.
- **Points against**
  1. No history of pleural effusion in the past.
  2. Left intercostal space fullness.
  3. Rapid progression of symptoms.
  4. Tracheal deviation

3.EMPYEMA

- **Points in favour**

1.Dull on percussion.

2.Reduced, vocal resonance and vocal fremitus.

- **Points against**

1.Non toxic

2.Fever not associated with chills and rigors

3.Absence of intercostal tenderness

4. MASSIVE CONSOLIDATION

- **Points in favour**
  1. Dullness on percussion
  2. History of cough with expectoration.
- **Points against**
  1. Non toxic
  2. Not an acute presentation
  2. Shifting of trachea to opposite side
  3. Absence of tubular breathing
  4. Involvement of whole of lung.

INVESTIGATIONS

1. Routine blood investigations
2. Chest X ray
3. Aspiration of pleural fluid
   -Nature of fluid
   -Biochemical
   -cytological
   -Special tests-Adenosine deaminase activity
4. Sputum examination
5. CT thorax
6. Pleural biopsy

TREATMENT

- Treat the cause
- Symptomatic treatment
  1. Medical management
  2. Thoracocentesis
     - Intercostal drainage
     - Thoracotomy
     - Rib resection

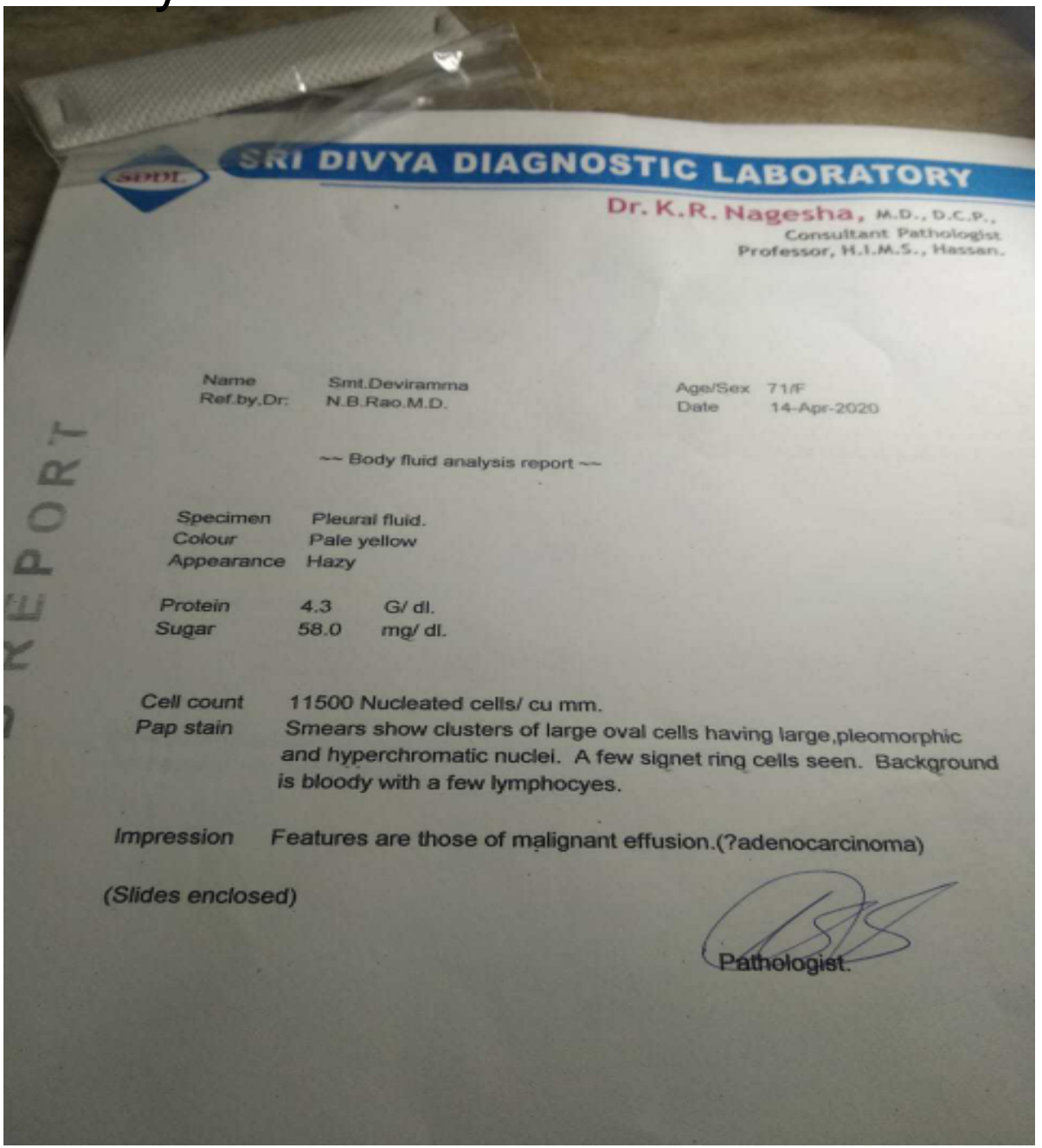
Chest X-ray PA view



???

? Causes of Pl effusion without mediastinal shift ??

Pleural fluid analysis



SRI DIVYA DIAGNOSTIC LABORATORY

Dr. K.R. Nagesha, M.D., D.C.P.,
Consultant Pathologist
Professor, H.I.M.S., Hassan.

REPORT

Name Smt.Deviramma
Ref.by.Dr: N.B.Rao.M.D.

Age/Sex 71/F
Date 14-Apr-2020

~~ Body fluid analysis report ~~

Specimen	Pleural fluid.	
Colour	Pale yellow	
Appearance	Hazy	
Protein	4.3	G/ dl.
Sugar	58.0	mg/ dl.

Cell count 11500 Nucleated cells/ cu mm.

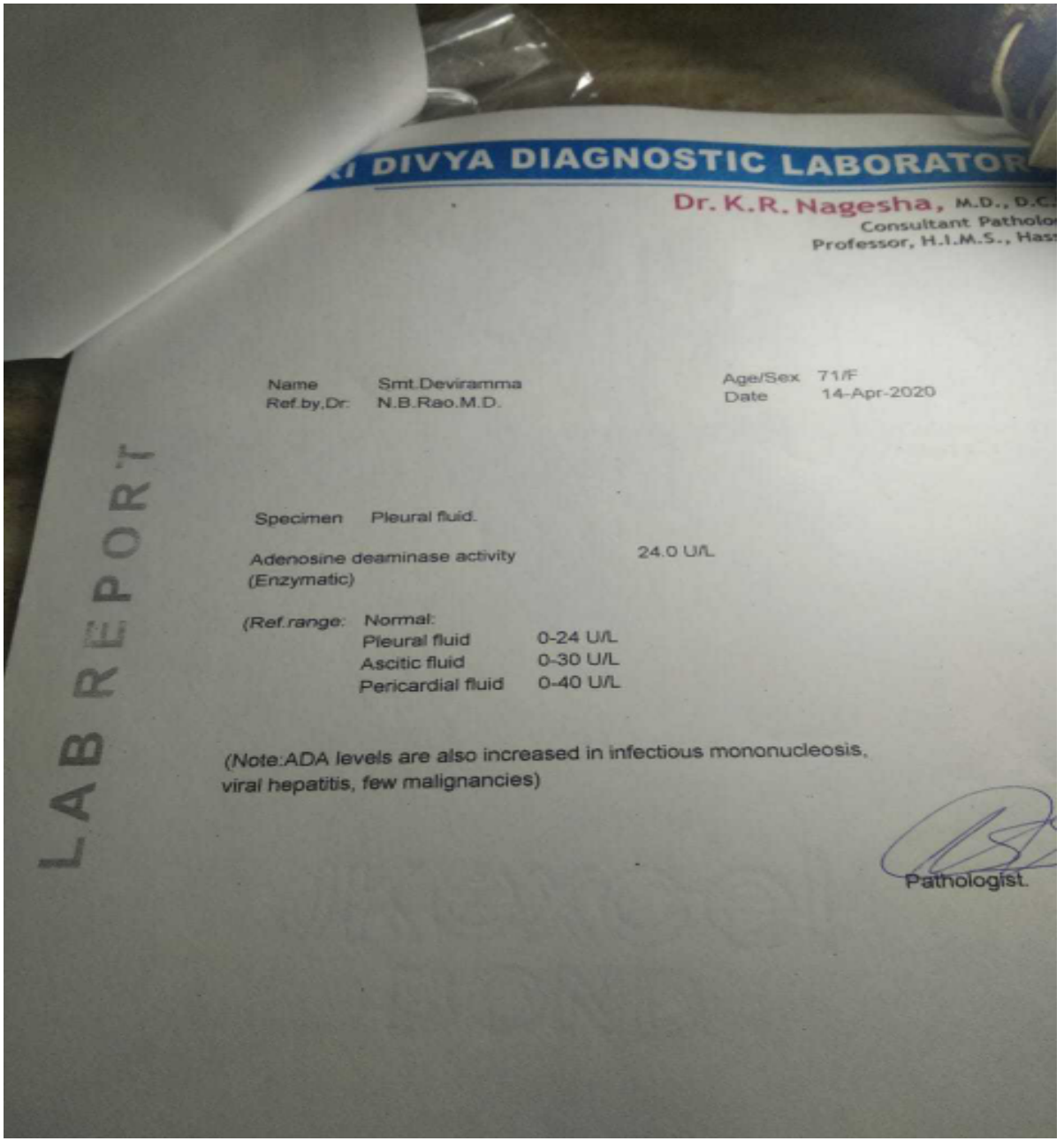
Pap stain Smears show clusters of large oval cells having large,pleomorphic and hyperchromatic nuclei. A few signet ring cells seen. Background is bloody with a few lymphocyes.

Impression Features are those of malignant effusion.(?adenocarcinoma)

(Slides enclosed)

Pathologist.

Pleural fluid analysis (continued)



I DIVYA DIAGNOSTIC LABORATOR

Dr. K.R. Nagesha, M.D., D.C.
Consultant Patholo
Professor, H.I.M.S., Has

LAB REPORT

Name Smt.Deviramma
Ref.by.Dr: N.B.Rao.M.D.

Age/Sex 71/F
Date 14-Apr-2020

Specimen Pleural fluid.

Adenosine deaminase activity (Enzymatic)	24.0 U/L

(Ref.range: Normal:

Pleural fluid	0-24 U/L
Ascitic fluid	0-30 U/L
Pericardial fluid	0-40 U/L

(Note:ADA levels are also increased in infectious mononucleosis, viral hepatitis, few malignancies)

Pathologist.

- Types of Pleural effusion :
  - Transudative
  - Exudative
  - Empyema
  - Hemorrhagic effusion
  - Chylous effusion

?? Mechanisms of Pleural Effusion

- ↑ Hydrostatic pressure
- ↓ Oncotic pressure
- ↑ Vascular permeability
- ↓ lymphatic drainage
- ↑Negative pressure in pleural space

? Causes of Pleural effusion

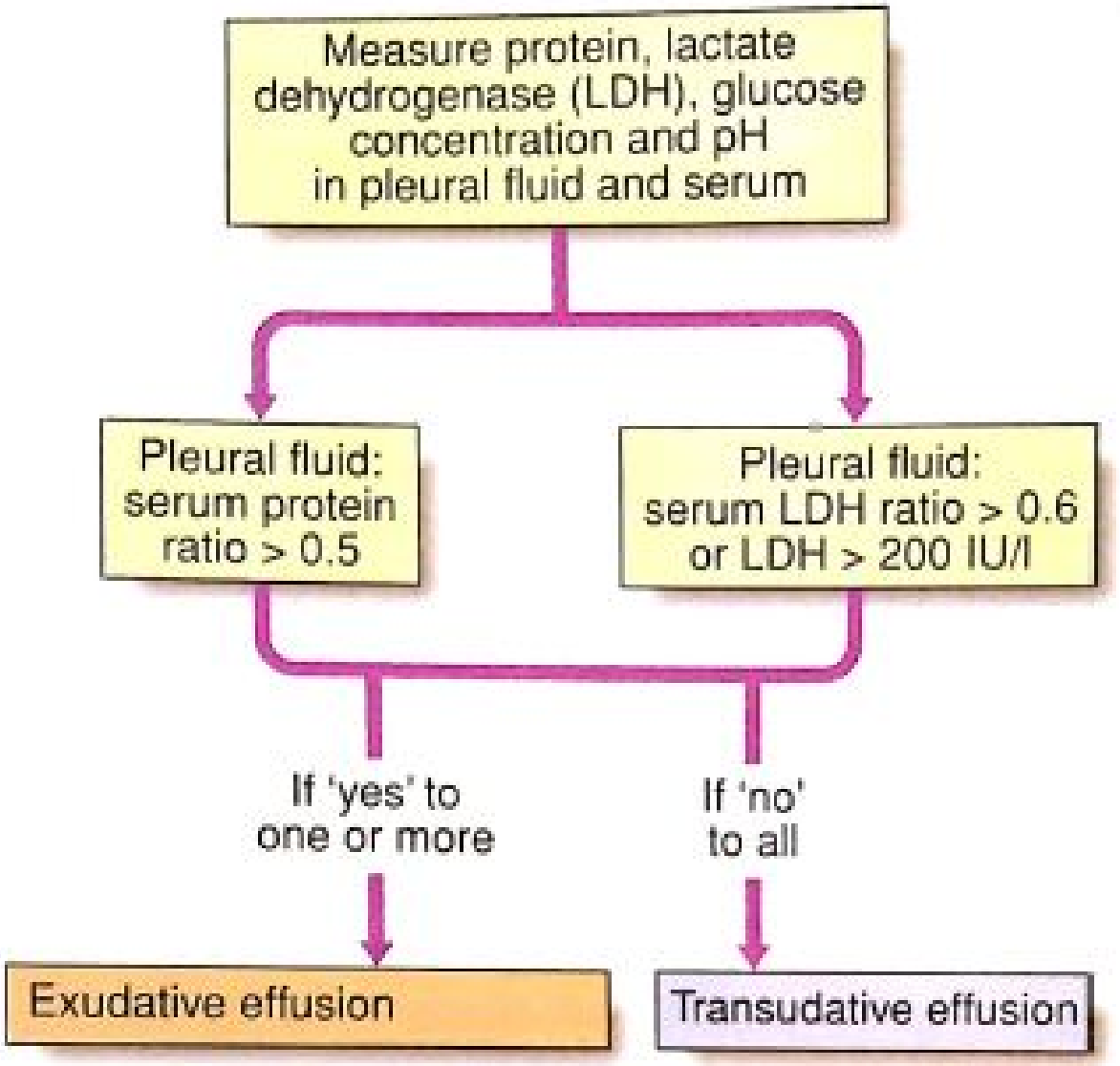
Common

- Pneumonia ('para-pneumonic effusion')
- Tuberculosis
- Pulmonary infarction
- Malignant disease
- Cardiac failure
- Subdiaphragmatic disorders (subphrenic abscess, pancreatitis etc.)

Uncommon

- Hypoproteinaemia (nephrotic syndrome, liver failure, malnutrition)
- Connective tissue diseases (particularly systemic lupus erythematosus and rheumatoid arthritis)
- Acute rheumatic fever
- Post-myocardial infarction syndrome
- Meigs' syndrome (ovarian tumour plus pleural effusion)
- Myxoedema
- Uraemia
- Asbestos-related benign pleural effusion

Light`s criteria ??



s:

ural fluid glucose very (< 1.4 mmol/l) in erculous and umatoid effusions

ural fluid cholesterol ially high (> 45 mg/dl)

normal or low (< 7.3); in empyema and very in oesophageal rupture

ylase high in effusions ociated with pancreatitis oesophageal rupture

udy or milky appearance > 4 g/l) indicates lous effusion

ural fluid complement els low in rheumatoid or E-related effusions

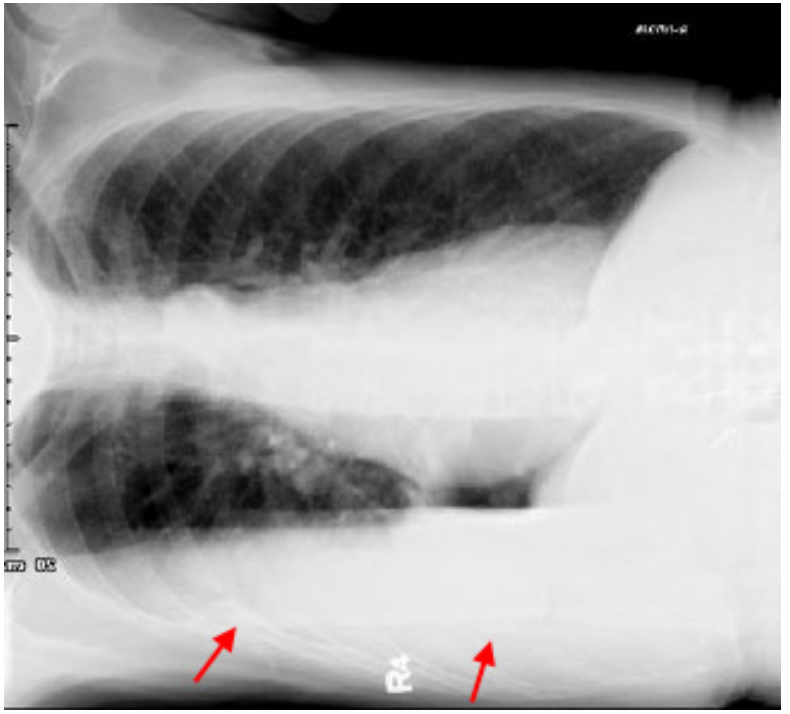
Notes:

- Pleural fluid glucose, amylase, cholesterol and pH all normal
- Causes: heart failure, liver failure, renal failure, thyroid failure, malnutrition
- Can occur in pulmonary infarction

- Rt sided Pleural effusion is more common?
- ?Quantity required for clinical detection of Pl effusion
- Signs of Pl effusion (Lt / Rt side)
- ? Bronchial BS in Pl effusion
- ? Lamellar effusion
- Indication for thoracocentesis in Rt Pl effusion?

CXR

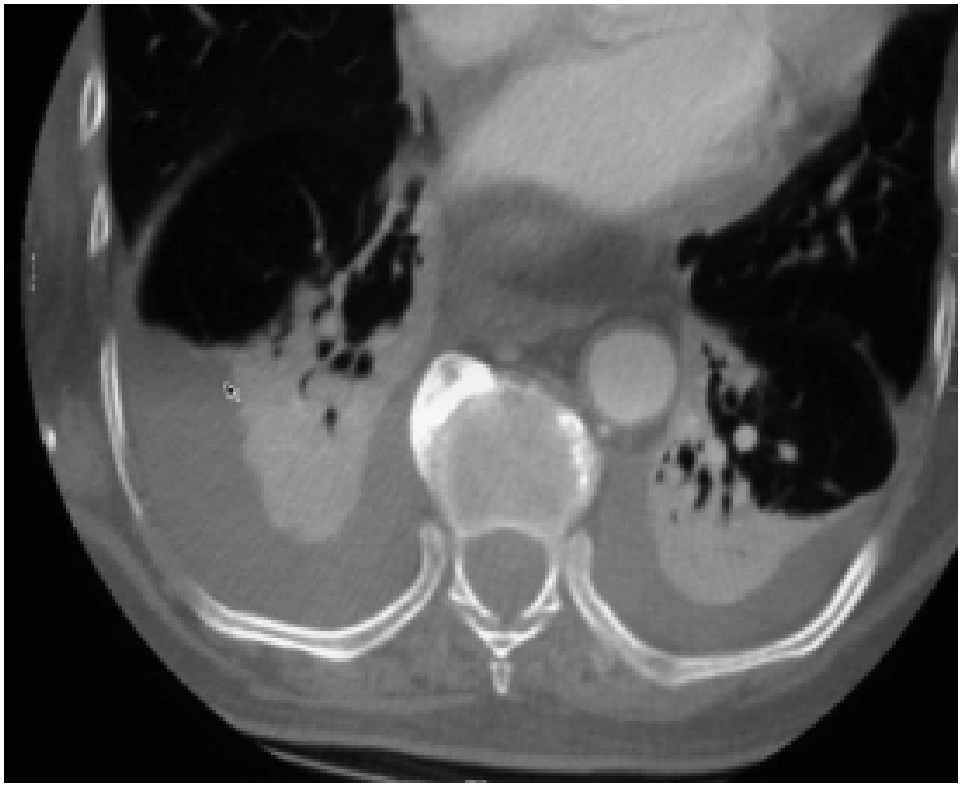


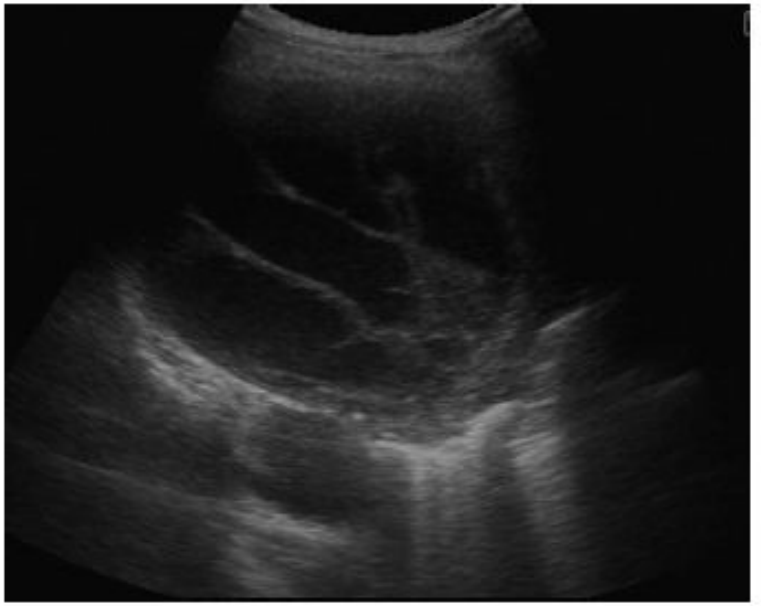


CXR- D /d

- Raised hemidiaphragm,
  - Hepatomegaly , Phrenic N palsy
- Collapse or consolidation
- Pleural thickening
  - Old TB / Empyema
- Poor radiographic technique

Role of CT Scan

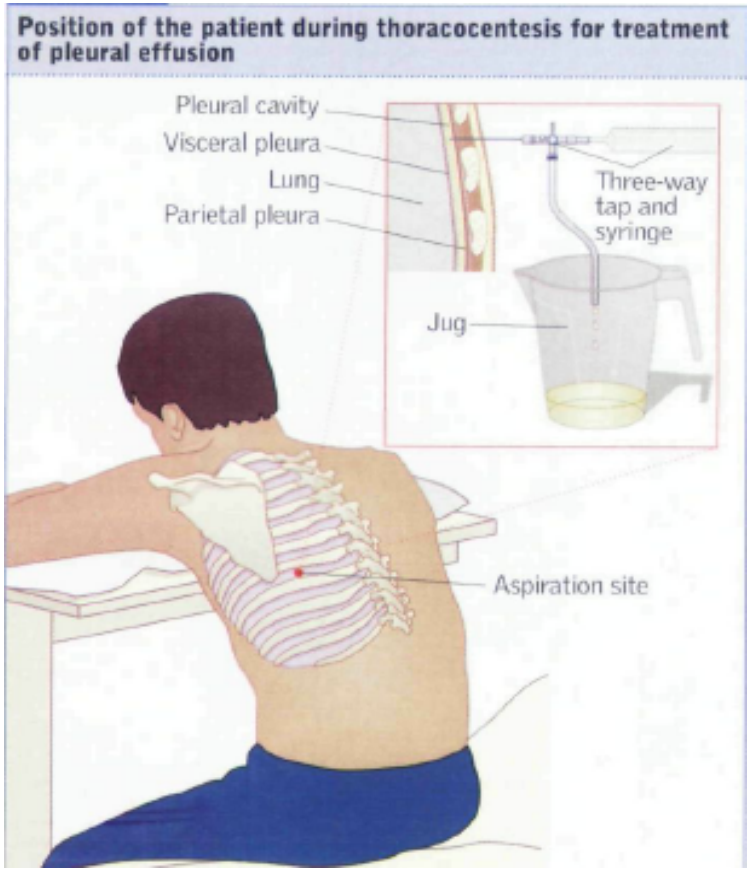




- Role of Ultrasound
  - Can help identify free vs. loculated effusions
  - Thoracentesis is facilitated by ultrasound guidance

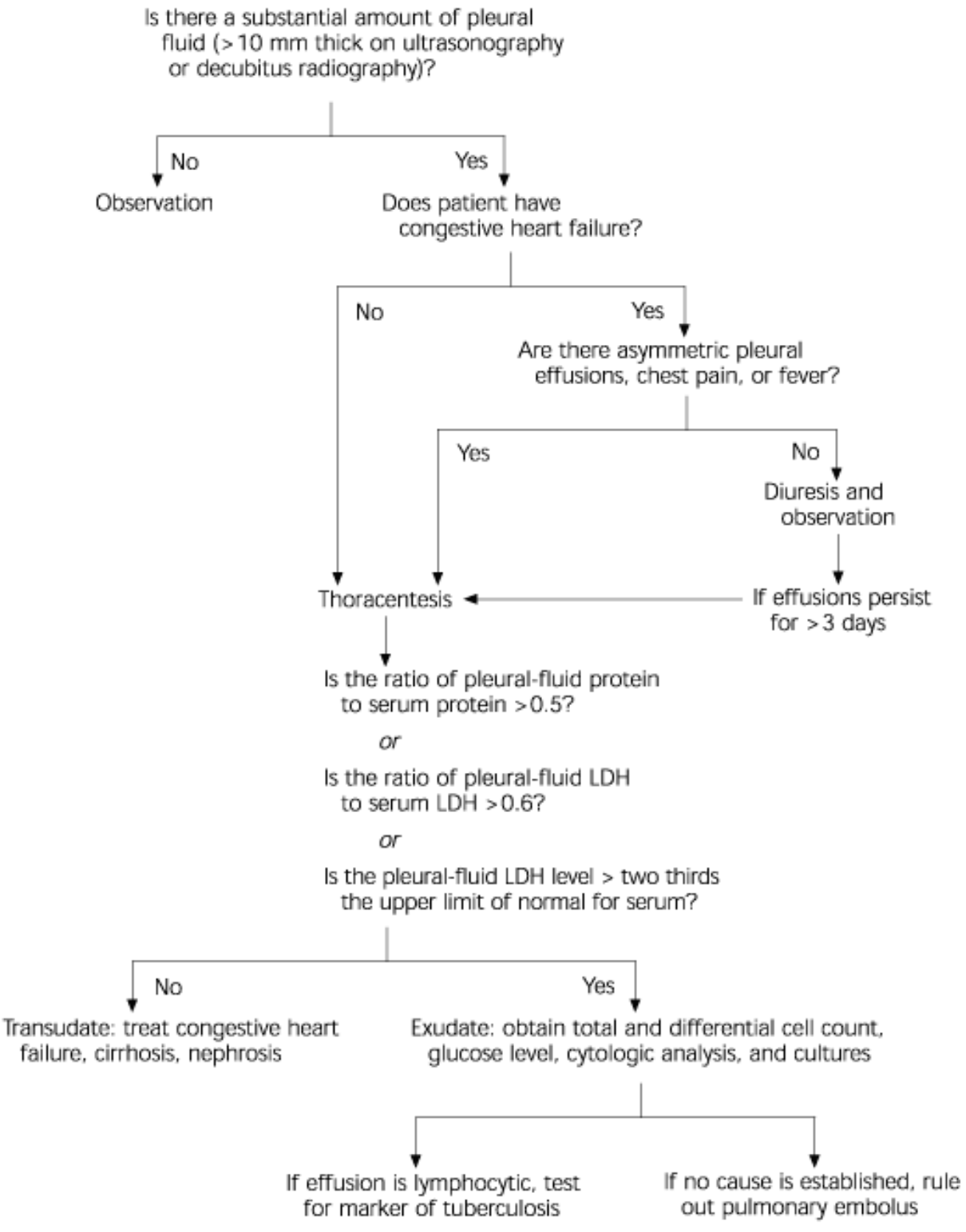
Indications for thoracentesis

- All Left sided effusions
- Pleural effusion of unknown etiology , with >10mm depth on lateral decubitus CXR
- Therapeutic
- Empyema
- Hydro-pneumothorax



Pleural fluid

- ? Bloody:
  - Hct< 1% not significant
  - 1-20%= CA, PE, Trauma
  - >50% serum Hct = hemothorax
- ? Cloudy
  - Triglyceride level >110mg% or Cholesterol > 60 mg% = chylothorax
- ? Foul smell
  - Infection? _ C/s; Gr staining..
- Amylase (Pl > Sr= Pacreatitic)
- PH < 7.2 = Empyema, TB, Ca..
- PH very Low PH = Eso rupture

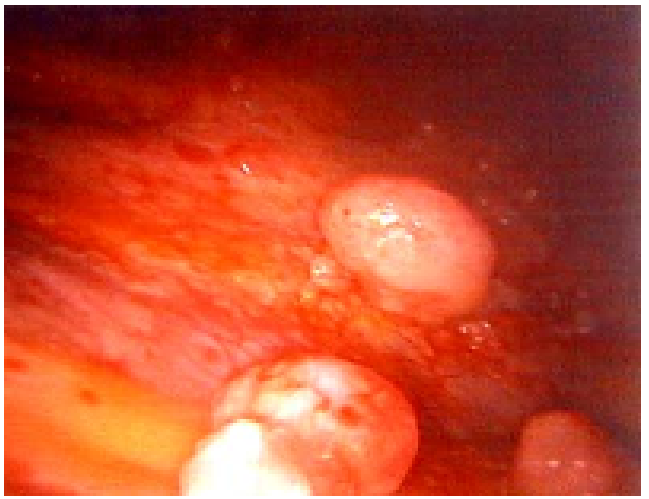

Is there a substantial amount of pleural fluid (>10 mm thick on ultrasonography or decubitus radiography)?
No
Yes
Observation
Does patient have congestive heart failure?
No
Yes
Are there asymmetric pleural effusions, chest pain, or fever?
Yes
No
Diuresis and observation
If effusions persist for >3 days
Thoracentesis
Is the ratio of pleural-fluid protein to serum protein >0.5?
or
Is the ratio of pleural-fluid LDH to serum LDH >0.6?
or
Is the pleural-fluid LDH level > two thirds the upper limit of normal for serum?
No
Yes
Transudate: treat congestive heart failure, cirrhosis, nephrosis
Exudate: obtain total and differential cell count, glucose level, cytologic analysis, and cultures
If effusion is lymphocytic, test for marker of tuberculosis
If no cause is established, rule out pulmonary embolus

- Pleural Biopsy
  - In evaluating for TB
  - Utility for CA (40-50% positive)
  - Sarcoid, fungal



- Thoracoscopy
  - Most helpful in evaluating for





? Indications for ICD

- Empyema
- Complicated parapneumonic effusion
- Hemothorax
- Malignant effusion

At the end it is my duty to thank

- Mrs.XYZ for her fullest cooperation during examination
- White Army and the teachers
- My friends for listening my case presentation.

Thank you all